



# A lightweight deep convolutional neural network model for skin cancer image classification

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## HIGHLIGHTS

- We proposed a new CNN is termed TurkerNet.
- The proposed TurkerNet is a lightweight CNN.
- TurkerNet achieved over 92% test accuracy on a public skin image dataset.

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## ABSTRACT

Deep learning models, particularly transformers and convolutional neural networks (CNNs), have been commonly used to achieve high classification accuracy for image data. Since introducing transformers, researchers have predominantly embraced these models to obtain impressive classification rates with novel approaches. In light of this scenario, we present a novel lightweight CNN called TurkerNet. Our primary objective is to attain a superior classification performance while minimizing the number of trainable parameters. TurkerNet comprises four essential components: the input block, residual bottleneck block, efficient block, and output block. To evaluate the performance of our proposed model, we conducted experiments using an open-access image dataset, specifically curated to include skin cancer images classified into two categories: benign and malignant. Our proposed (TurkerNet) model achieved a remarkable testing accuracy of 92.12% on this public dataset. Our model performed better than state-of-the-art techniques developed for automated skin cancer detection. Moreover, our proposed TurkerNet is a lightweight model. In this aspect, the presented TurkerNet is highly accurate with low trainable parameters.

## 1. Introduction

Ultraviolet (UV) rays can cause the development of skin cancer, and they are categorized into UVA, UVB, and UVC. UVC rays are mostly absorbed by the atmosphere and do not reach the skin's surface. Both UVA and UVB rays can increase the risk of skin cancer. UVB rays are particularly associated with directly causing melanoma skin cancer. The damaging wavelength range for UVA rays is 320–400 nanometers (nm), while for UVB rays, it is 290–320 nm. The damaged wavelength of these UVs is 320–400 nanometers (nm) for A and 290–320 B-type UV rays. Skin cancer arises from the aberrant proliferation of pigmentation and skin cell growth, commonly known as malignant melanoma [1].

Detecting disorder in melanoma cells during the initial stages is crucial in advising dermatologists to recommend surgical removal of the affected skin area to prevent further spread [2,3]. Accurate identification and confirmation of malignant melanoma in its early stages is important due to the significantly higher likelihood of successful treatment. Unfortunately, conventional diagnostic methods do not facilitate straightforward skin cancer detection [4]. Melanoma, one of the most dangerous types of cancer, falls within the category of lethal malignancies [5]. Its genesis lies in the deoxyribonucleic acid (DNA) of skin cells, where the deoxyribonucleic acid incites mutations that rapidly propagate, necessitating early detection to impede its dissemination and minimize infection in unaffected skin regions [6].

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Automatic diagnosis and diagnostic systems based on machine learning techniques have found extensive use in various fields, including medical applications [7–9]. These applications have significantly reduced the workload of experts. Additionally, decision support systems are being developed for preliminary diagnosis in areas where experts are unavailable [10,11]. We have reviewed some state-of-art models for automatic skin cancer detection and these are explained below.

Huang et al. [12] developed a skin cancer classification method. Their method was based on YOLOv5 and hyperspectral imaging. They used a total of 1854 images for three classes (basal cell carcinoma, seborrheic keratosis, and squamous cell carcinoma) and achieved an accuracy of 79.20%. Teodoro et al. [13] proposed a skin cancer detection approach with a region of interest-based attention mechanism and generative adversarial network. By applying these methods, they attained an accuracy of 97.90% for 58989 images. Dandu et al. [14] focused on transfer learning for skin cancer classification. They used the International Skin Imaging Collaboration 2018 dataset (10000 images). In their presented method, the accuracy value was obtained as 90.96%. Balaha and Hassan [15] presented a skin cancer detection method using a sparrow search algorithm and deep transfer learning. They used five datasets (ISIC 2019–2020, HAM10000, Skin diseases image dataset, Skin cancer segmentation/classification dataset, and PH2). They attained accuracies of 98.27%, 98.83%, 85.87%, 94.16%, and 94.75% for ISIC 2019–2020, HAM10000, skin diseases image dataset, skin cancer segmentation/classification dataset, and PH2, respectively. Mukadam and Patil [16] developed a classification method for skin cancer with CNN and a super-resolution generative adversarial network. They used the HAM10000 dataset and obtained an accuracy of 98.77% with data augmentation. Obayya et al. [17] used the Henry gas solubility optimization algorithm for skin cancer detection. In their proposed feature extraction method, ISIC2017 and HAM10000 datasets were used. They achieved 94.50% and 99.74% accuracy for the ISIC2017 and HAM10000 datasets, respectively. Javid et al. [18] proposed a melanoma skin cancer classification method using deep ensemble learning techniques. The accuracy rate was 93.00% using the ISIC dataset (10600 images for benign-malignant classes). Keerthana et al. [19] developed a skin cancer detection method with hybrid convolutional neural networks. They used the ISBI 2016 dataset (1279 images) and attained an accuracy of 86.23%. Kumar et al. [20] used a deep neural networking framework to detect skin cancer. Hence, they used Kaggle Re-Snet50 datasets and reported an accuracy of 93.26%. He et al. [21] employed CNN to predict skin cancer disease. He et al. used a seven-point checklist dataset. Their proposed method obtained the average accuracy rate as 76.80%. Aishwarya et al. [22] used the YOLO deep neural network to detect skin cancer disease using the ISIC dataset (1460 images). They attained an average accuracy rate of 98.06%. Rao et al. [23] used a method to categorize different types of skin cancer lesions. They used deep learning architectures and achieved an accuracy of 87.00% with 2637 images.

### 1.1. Motivation and the proposed model

Our primary objective is to contribute to the field of convolutional neural networks (CNNs) by introducing a novel deep image classification model. Hence, we presented TurkerNet, a simple yet highly effective image classification model. Our work encompasses two key goals: (i) achieving superior classification performance while (ii) minimizing the number of trainable parameters.

To develop TurkerNet, we conducted an extensive analysis of the structural characteristics of renowned models such as ResNet [24], MobileNet [25], and EfficientNet [26]. Inspired by MobileNet, we devised an input block employing a bottleneck block. Subsequently, we designed a residual block (MBConv6) and an effective block (squeeze and exception block). To reduce dimensionality, we employed maximum pooling. Finally, we incorporated an output block to obtain classification results. The graphical representation of our model is

shown in Fig. 1.

### 1.2. Novelties and contributions

We have presented a new CNN (TurkerNet) and the innovations and contributions of this CNN are:

- In the existing literature, there are various CNN models. However, after proposing transformers for computer vision, most researchers have used transformers in their models to get high classification performance. This situation causes a stagnation in the CNN's development [27–29]. To handle this stagnation problem, we have presented a new CNN model by proposing a new block which is the combination of a residual and squeeze-and-excitation block. Using this block, we have proposed a new CNN called TurkerNet.
- Our proposed TurkerNet is a lightweight CNN since it has less than 10 million learnable parameters. The proposed TurkerNet is an alternative model to MobileNet, SqueezeNet, ShuffleNet, etc.
- The proposed TurkerNet attained 92.12% of test classification accuracy on the publicly available dataset.

## 2. Dataset

In this study, we employed an openly accessible [30] skin tumor image dataset readily available on Kaggle. Our primary focus was to obtain comparative results utilizing this specific dataset. The dataset consists of two main sections: the training and testing sets, which showcase distinct classes, namely benign and malignant. The details of the number of images used to develop the proposed model are shown in Table 1.

## 3. The proposed CNN

This paper presents TurkerNet, a novel deep learning network that employs four distinct blocks to achieve accurate classification results. We proposed a lightweight convolutional neural network (CNN) architecture as the foundation of TurkerNet, enabling its use as a lightweight model. Furthermore, our architecture can be scaled up to accommodate large-scale datasets, enabling effective classification using extensive datasets.

TurkerNet comprises 97 layers, which are organized into five key blocks: the input block, the residual block, the efficient block, the dimension reduction block (conv. block, as depicted in Fig. 1), and the output block. The input block is designed to mimic the structure of MobileNet, utilizing a bottleneck block. Inspired by the literature's findings on the superior classification capabilities of ResNets, which address the issue of vanishing gradients through residual blocks, we incorporated residual blocks into our architecture. To achieve high classification performance with a reduced number of trainable parameters, we drew inspiration from the EfficientNet architecture and introduced an efficient block. The output block follows a similar structure to that of other CNNs. Additionally, we utilized a convolution block to decrease the dimensionality of the generated features. Fig. 2 provides a visual demonstration of the employed blocks.

The transition list of the proposed TurkerNet is also listed in Table 2.

As indicated in Table 2, TurkerNet does not include any duplicated blocks. By incorporating block repetition, we can extend this model to support the training of large-scale datasets, enabling its applicability to handle substantial data.

Moreover, we have given the mathematical expressions of the presented TurkerNet and these are given below.

For the input block.

$$I_1 = \text{Conv}_{3 \times 3, \quad 32, \text{Stride}=2}(I) \quad (1)$$

$$I_2 = \text{Conv}_{3 \times 3, \quad 16, \text{Stride}=2}(I_1) \quad (2)$$

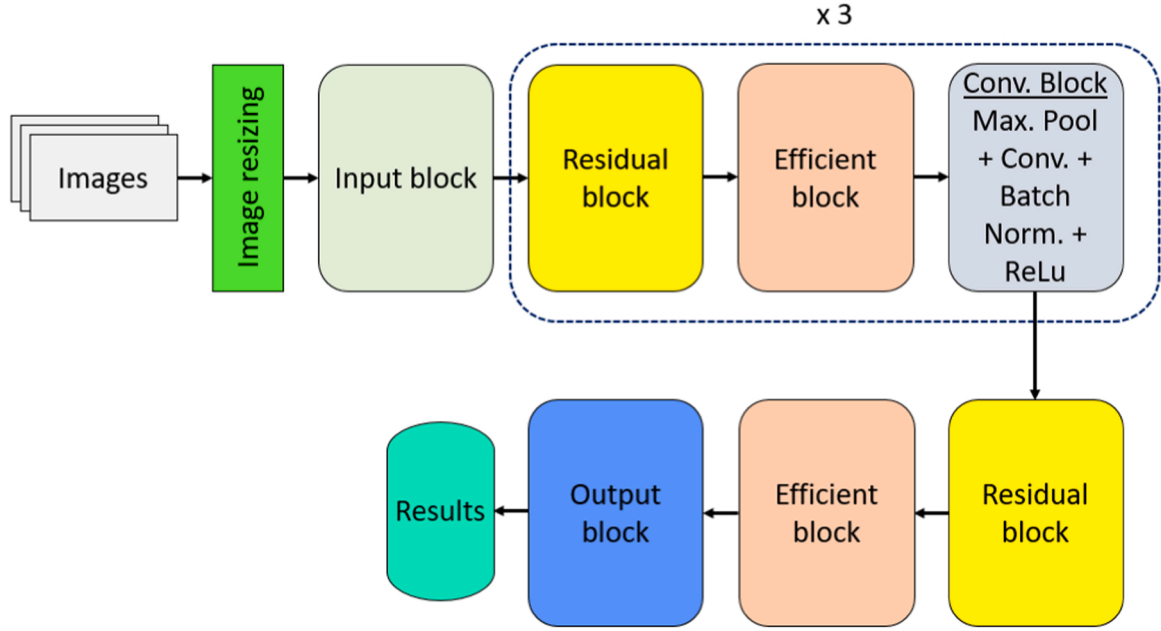


Fig. 1. Graphical representation of the proposed TurkerNet. Efficient block is a squeeze exception (SE) block.

Table 1

Details of the number of images used to develop the proposed model.

| Class     | Train | Test | Total |
|-----------|-------|------|-------|
| Malignant | 1197  | 300  | 1497  |
| Benign    | 1440  | 360  | 1800  |
| Total     | 2637  | 660  | 3297  |

It may be noted from Table 1 that 2637 train and 660 test images in this dataset.

$$I_3 = \text{Conv}_{1 \times 1, 64}(I_2) \quad (3)$$

Herein,  $I$ : Image with a size of  $224 \times 224 \times 3$ ,  $\text{Conv}$ : convolution operator and we have demonstrated the parameters of the convolutions which are the number of filters and stride. Moreover, this function contains batch normalization and ReLU operators. Herein, the output is  $I_3$  and the size of this output is equal to  $56 \times 56 \times 64$ .

To create the main block, we have used a residual block and the mathematical depiction of this block is given below.

$$R_n = \text{Conv}_{1 \times 1, C} \left( \text{Conv}_{3 \times 3, \frac{C}{4}} (\text{Conv}_{1 \times 1, C} (R_{n-1})) \right) + R_{n-1} \quad (4)$$

Herein,  $R$ : residual output,  $C$ : number of filters.

Moreover, we adapted the SE block to the above residual block and it is given below.

$$\text{Out}_n = \text{Conv}_{1 \times 1, C} \left( \text{Conv}_{1 \times 1, \frac{C}{4}} (\text{GAP}(R_n)) \right) \times R_n \quad (5)$$

Herein,  $\text{Out}$ : the outcome of the main block and  $\text{GAP}$ : global average pooling.

The formulas of the dimension reduction block are:

$$D_1 = \text{MaxPool}_{3 \times 3, C, \text{Stride}=2}(\text{Out}_n) \quad (6)$$

$$D = \text{Conv}_{1 \times 1, 2C}(D_1) \quad (7)$$

Where  $\text{MaxPool}$ : the maximum pooling function.

The last block is the output block.

$$\text{Out} = \text{SoftMax}(\text{FC}(\text{GAP}(\text{O}_{\text{last}}))) \quad (8)$$

Here,  $\text{Out}$ : the final output,  $\text{FC}$ : fully connected layer.

Moreover, the general mathematical definition of the presented model is defined in Eq. (9).

$$R = \{1, 1, 1, 1\}, \quad C = \{64, 128, 256, 512\} \quad (9)$$

Where  $R$ : number of repetitions and  $C$ : number of filters.

#### 4. Experiment

Our experiments were conducted on a personal computer (PC) with a modest configuration. The PC featured an NVidia 2060 graphical processing unit (GPU) card, 32 GB of memory, and a 3.6 GHz processor, and it ran on the Windows 11 operating system. To construct the proposed TurkerNet model, we used the MATLAB Deep Network Designer, a valuable tool for development.

Throughout the training process, we adopted a training and validation split ratio of 70:30. The training options employed in this study are as follows: Solver: sgd, initial learning rate: 0.01, mini-batch size: 32, number of epochs: 30, and hardware: graphical processing unit. Fig. 3 shows the training and validation curves, offering valuable insights into the model's progression during training.

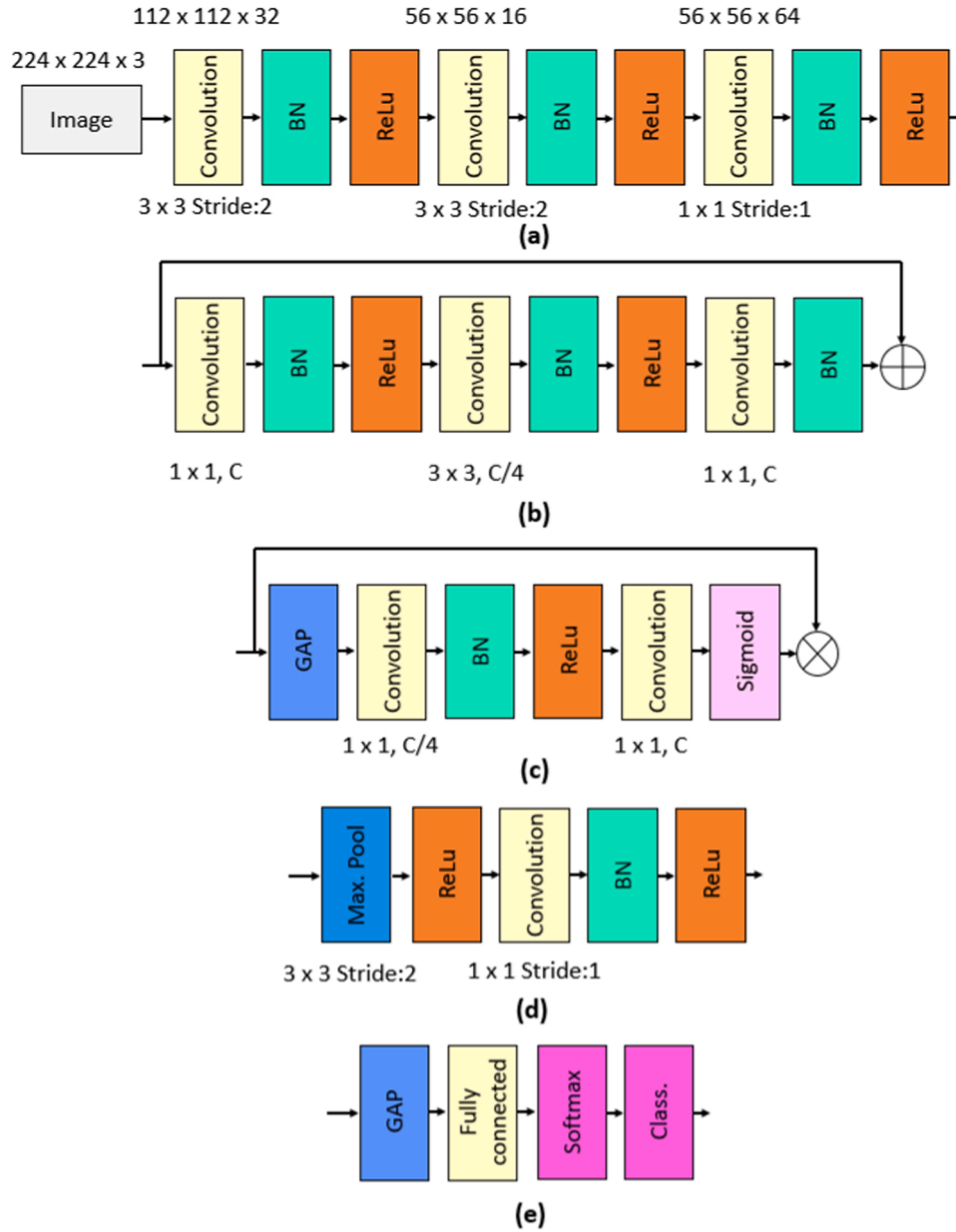
Our proposed model achieved a validation accuracy of 89.07% and a perfect training accuracy of 100%. Furthermore, we computed the testing accuracy of our model and presented the confusion matrix shown in Fig. 4.

Fig. 4 depicts the TurkerNet model obtained a commendable testing accuracy of 92.12%. Furthermore, we meticulously calculated the recall, precision, and F1-score of the test results for the TurkerNet model. The detailed outcomes of these metrics are diligently documented in Table 3.

Table 3 illustrates that our proposed method has attained satisfactory performance across all performance metrics.

#### 5. Discussions

We have presented a new CNN model and this CNN model is named TurkerNet. TurkerNet is a lightweight CNN and we aim to get high classification performance with lower trainable parameters. Therefore, we have compared this model to the well-known lightweight CNN which



**Fig. 2.** Details of blocks used in the presented TurkerNet. (a) Input, (b) residual, (c) efficient/squeeze-and-excitation (SE) block, (d) decomposition and (e) output blocks.

is MobileNetV2. We have conducted a comprehensive performance comparison between our model and MobileNetV2, presenting the comparative results in Table 4.

As illustrated in Table 4, our proposed TurkerNet exhibited a classification accuracy that surpasses MobileNetV2 by over 5% for the skin tumor image dataset under consideration.

We have compared the performance of our proposed TurkerNet to three commonly used lightweight CNNs: SqueezeNet, ShuffleNet, and EfficientNetb0. The comparative results are shown in Fig. 5.

As seen in Fig. 5, the proposed model achieved higher classification performance than other CNNs (SqueezeNet, ShuffleNet, EfficientNetb0). The proposed TurkerNet achieved over 10% higher test classification accuracy than SqueezeNet. EfficientNetb0 was the closest competitor, and our proposed TurkerNet achieved over 5.04% higher test classification accuracy than EfficientNetb0.

The comparison of our proposed method with other state-of-the-art methods is given in Table 5.

As seen in Table 5, the proposed method is effective for skin cancer

detection. It achieved the most successful results compared to other methods.

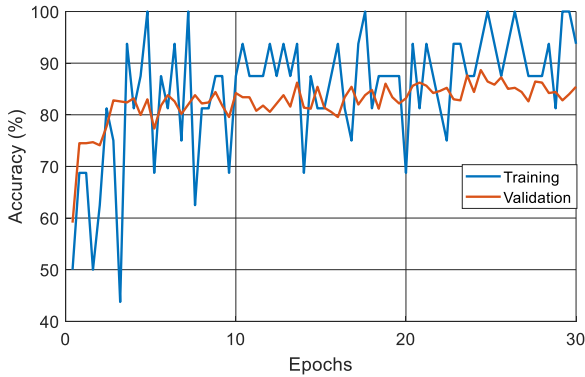
The key findings from our work are outlined below:

- We introduced a novel generation deep convolutional neural network (CNN) architecture by incorporating mobile inverted bottleneck (MBConv4) and SE 0.25 blocks.
- Our proposed model was evaluated on a publicly available dataset of skin image cancer.
- TurkerNet comprised 97 layers and approximately 2.2 million trainable parameters.
- Notably, our model boasts approximately 2.2 million parameters, further highlighting its efficiency and effectiveness in image classification tasks. These compelling results undeniably establish our proposed CNN, TurkerNet, as an exemplary model for image classification endeavors.
- Our model achieved superior classification performance compared to MobileNetV2 when applied to this specific dataset.

**Table 2**

Transition list of the proposed TurkerNet.

| Stage | Block         | Input     | Operation                                                                                                                | Output    |
|-------|---------------|-----------|--------------------------------------------------------------------------------------------------------------------------|-----------|
| 1     | Input         | 224×224×3 | $\begin{bmatrix} 3 \times 3, 32, \text{Stride} = 2 \\ 3 \times 3, 16, \text{Stride} = 2 \\ 1 \times 1, 64 \end{bmatrix}$ | 56×56×64  |
| 2     | Residual      | 56×56×64  | $\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 16 \\ 1 \times 1, 64 \end{bmatrix}$                                       | 56×56×64  |
| 3     | Efficient/SE  | 56×56×64  | $\begin{bmatrix} 1 \times 1, 16 \\ 1 \times 1, 64 \end{bmatrix}$                                                         | 56×56×64  |
| 4     | Decomposition | 56×56×64  | $3 \times 3, \text{maxpool}, \text{Stride} = 2$ [1 × 1, 128]                                                             | 28×28×128 |
| 5     | Residual      | 28×28×128 | $\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 32 \\ 1 \times 1, 128 \end{bmatrix}$                                     | 28×28×128 |
| 6     | Efficient     | 28×28×128 | $\begin{bmatrix} 1 \times 1, 32 \\ 1 \times 1, 128 \end{bmatrix}$                                                        | 28×28×128 |
| 7     | Decomposition | 28×28×128 | $3 \times 3, \text{maxpool}, \text{Stride} = 2$ [1 × 1, 256]                                                             | 14×14×256 |
| 8     | Residual      | 14×14×256 | $\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix}$                                     | 14×14×256 |
| 9     | Efficient     | 14×14×256 | $\begin{bmatrix} 1 \times 1, 64 \\ 1 \times 1, 256 \end{bmatrix}$                                                        | 14×14×256 |
| 10    | Decomposition | 14×14×256 | $3 \times 3, \text{maxpool}, \text{Stride} = 2$ [1 × 1, 512]                                                             | 7×7×512   |
| 11    | Residual      | 7×7×512   | $\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix}$                                    | 7×7×512   |
| 12    | Efficient     | 7×7×512   | $\begin{bmatrix} 1 \times 1, 128 \\ 1 \times 1, 512 \end{bmatrix}$                                                       | 7×7×512   |
| 13    | Output        | 7×7×512   | Global average pooling, fully connected, Softmax.                                                                        | 1×1       |

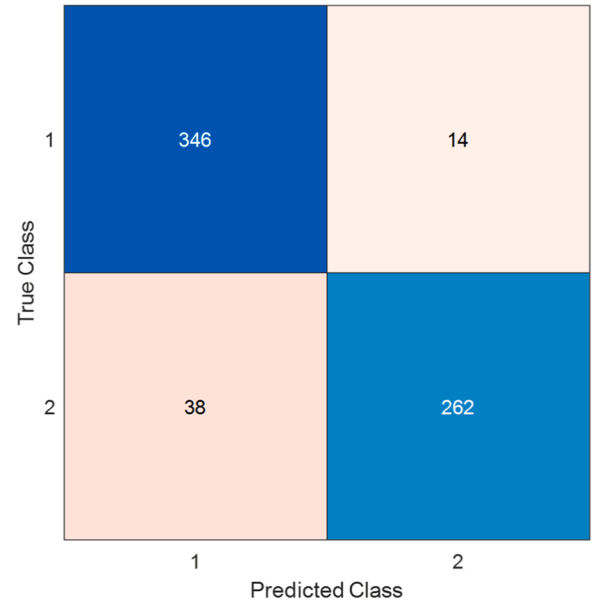
**Fig. 3.** Training and validation curves obtained for the proposed TurkerNet developed for automated skin cancer detection.

- By introducing TurkerNet, we contributed to the field of lightweight CNN research.
- However, it is worth noting that our model's performance has yet to be evaluated on larger datasets, such as ImageNet1k and ImageNetReal, which remains a limitation of our study.

### 5.1. Ablation

The main limitation of the proposed model is that TurkerNet was only tested on the skin cancer image dataset. Our primary objective is to achieve high classification performance on this challenging dataset, a commonly used and imbalanced skin cancer image dataset. Therefore, we proposed TurkerNet to attain high classification performance on this specific dataset. To address this problem, we used an additional dataset—the blood image dataset—which is also publicly available and can be downloaded from Kaggle [<https://www.kaggle.com/datasets/uncles-amulus/blood-cells-image-dataset>]. This dataset comprises 8 classes that represent blood cells. We used the same settings, and the computed training and validation curves are presented in Fig. 6.

Fig. 6 depicts that TurkerNet achieved 97.38% validation accuracy

**Fig. 4.** Confusion matrix obtained using the test data. \* 1; benign, 2; malignant.**Table 3**

Summary of results (%) obtained for different classes.

| Class              | Recall | Precision | F1-score |
|--------------------|--------|-----------|----------|
| Benign             | 96.11  | 90.10     | 93.01    |
| Malignant          | 87.33  | 94.93     | 90.97    |
| Unweighted average | 91.72  | 92.52     | 91.99    |

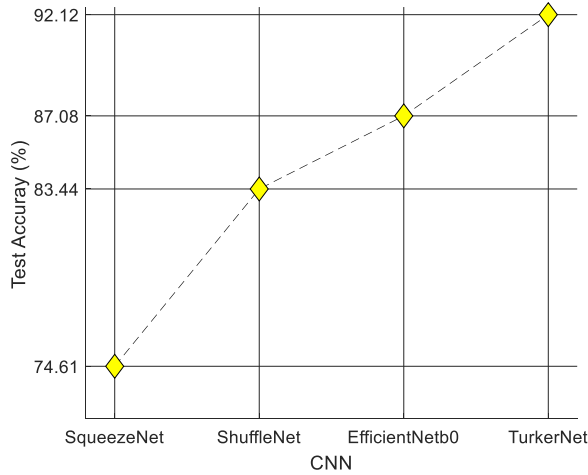
and 100% training accuracy on the blood image dataset. Additionally, TurkerNet reached 96.94% test accuracy on the blood image dataset. These results indicate that our model performs well on large image datasets and possesses a strong general classification ability.

To show the effectiveness of the presented TurkerNet, we have used

**Table 4**

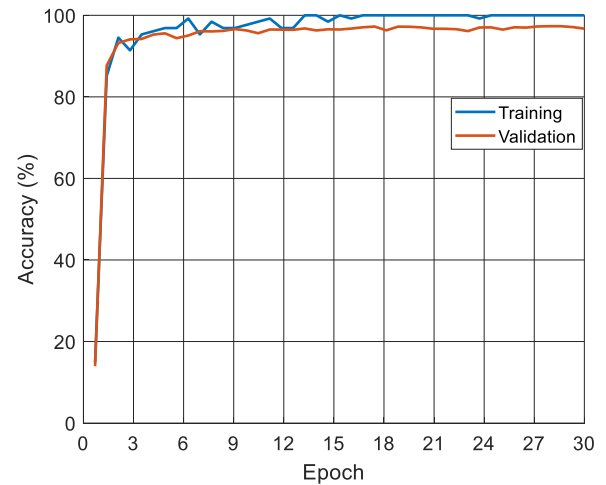
Comparison of results obtained and model details.

| Model       | Accuracy(%) | Number of parameters | Number of layers |
|-------------|-------------|----------------------|------------------|
| MobileNetV2 | 86.53%      | 2.2 M                | 154              |
| TurkerNet   | 92.12%      | 2.2 M                | 97               |

**Fig. 5.** Performance of the proposed TurkerNet compared to three commonly used lightweight CNNs.**Table 5**

Comparison of our proposed method with other state-of-the-art methods using the same dataset.

| Year              | Study                       | Method(s)                                                      | Classifier     | Accuracy (%) |
|-------------------|-----------------------------|----------------------------------------------------------------|----------------|--------------|
| 2021              | Kharisudin et al. [31]      | CNN                                                            | Softmax        | 90.00        |
| 2021              | Abdar et al. [32]           | Binary residual feature fusion                                 | Softmax        | 89.24        |
| 2021              | Abdar et al. [33]           | Three-way decision-based Bayesian deep learning                | Softmax        | 88.95        |
| 2021              | Gupta et al. [34]           | Inceptionv3                                                    | Neural network | 83.20        |
| 2022              | Anand et al. [35]           | VGG16                                                          | Softmax        | 89.09        |
| 2022              | Baygin et al. [36]          | Ensemble DarkNet, neighborhood component analysis              | SVM            | 91.54        |
| 2022              | Alfi et al. [37]            | CNN                                                            | Softmax        | 92.00        |
| 2022              | Salian and Sawarkar [38]    | EfficientNetB3                                                 | Softmax        | 87.12        |
| 2023              | Ramya and Sathiyabhama [39] | Enhanced genetic algorithm, extreme learning machine, ResNet50 | SVM            | 89.19        |
| 2023              | Hussein et al. [40]         | ResNet-18                                                      | Softmax        | 89.39        |
| 2023              | Shekar and Hailu [41]       | DenseNet-169, local binary pattern                             | Random forest  | 89.70        |
| 2023              | Nivyashree and Pramila [42] | CNN                                                            | Softmax        | 80.80        |
| 2023              | Ghosh et al. [43]           | CNN-based two-phase evolutionary framework                     | Softmax        | 83.40        |
| 2024              | Maurya et al. [44]          | CNN                                                            | XGBoost        | 87.88        |
| 2024              | Angelina and Ulfitria [45]  | VGG16                                                          | Softmax        | 83.75        |
| <b>Our method</b> |                             | <b>TurkerNet</b>                                               | <b>Softmax</b> | <b>92.12</b> |

**Fig. 6.** The training and validation accuracy of the proposed TurkerNet for the blood image dataset.

ablation studies and have defined three cases and these cases are:

**Case 1.** In this case, we used a ResNet50 block instead of our residual block.

**Case 2.** In this case, we have removed the SE block from our presented block.

**Case 3.** This case represents the TurkerNet.

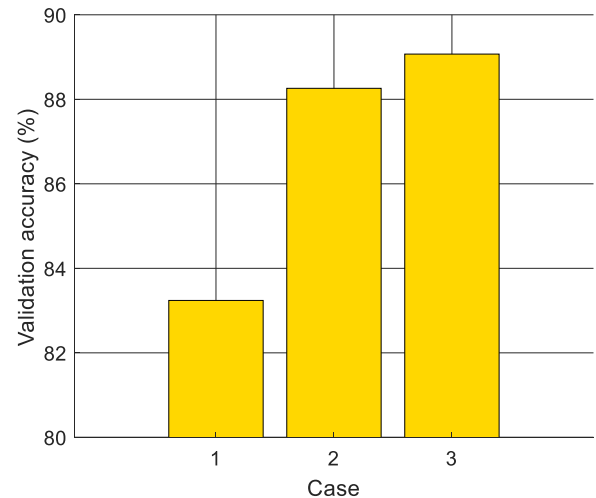
We have demonstrated the validation accuracies of these cases in Fig. 7.

Fig. 7 depicts that TurkerNet achieved the best results among the configurations tested. Therefore, we utilized this specific combination of components to create the proposed TurkerNet.

## 6. Conclusions

Our proposed deep learning network, TurkerNet, demonstrated high classification performance in image classification, especially for skin tumor images. By incorporating innovative architectural elements such as the mobile inverted bottleneck (MBConv4) and SE 0.25 blocks, TurkerNet achieved a superior test classification accuracy of 92.12% compared to existing models like MobileNetV2.

The primary focus of the model is to accurately classify skin images into benign (non-cancerous) and malignant (cancerous) categories. We

**Fig. 7.** The validation accuracies of the defined cases.



achieved 92.12% accuracy for this task on a publicly available dataset. This high level of accuracy indicates that TurkerNet is highly effective in distinguishing between benign and malignant skin cancer images, making it an important model for automatic skin cancer detection rather than broad skin image classification.

With 97 layers and approximately 2.2 million trainable parameters, TurkerNet yielded high performance and low computational complexity.

Our work contributed to the advancement of lightweight CNNs, showcasing the potential for improved classification performance while minimizing model complexity. However, it is important to note that further evaluation of TurkerNet on larger datasets, such as ImageNet1k and ImageNetReal, would provide valuable insights into its generalization capabilities. Since we tested the proposed model using limited datasets and we have not proposed scalable versions of the presented TurkerNet like ConvNeXt.

TurkerNet is a successful image classification model, and this CNN achieved high classification performance on two biomedical image datasets. These results and findings unequivocally demonstrate that the proposed TurkerNet can be used to solve other image classification tasks.

### Institutional review board statement

Not applicable.

### Informed consent statement

Not applicable.

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### CRedit authorship contribution statement

**Rajendra U. Acharya:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Turker Tuncer:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Prabal Datta Barua:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ilknur Tuncer:** Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sengul Dogan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

The skin cancer image dataset can be downloaded from [30]

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